Simran Christy Serrao

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EDUCATION

Birla Institute of Technology and Science (BITS) Pilani, KK Birla Goa Campus Oct 2022 – Present
B.E. (Hons.) Computer Science; M.Sc. (Hons.) Biological Sciences CGPA: 8.47/10
In-Major GPA (Biological Sciences): 9.82/10

AWARDS AND FUNDING

- **Khorana Program for Scholars** *UVRI at Stanford Medicine* Jun 2025 - Aug 2025
  A collaborative initiative between the Department of Biotechnology (Government of India), WINStep Forward, and the Indo-U.S. Science and Technology Forum that provides Indian students with 10 to 12 weeks of fully funded summer research internships at premier U.S. universities.
- **Merit Scholarship Recipient (Top 3%)** at BITS Pilani KK Birla Goa Campus. Jun 2024

MASTER'S THESIS (ONGOING: JUNE 2026 - JUNE 2027)

An Analysis of Social Network Dynamics among Indian STEM Undergraduates: Implications for Targeted Empathy Interventions
Supervisors: Dr. Aswini S., Dr. Indrani Talukdar, Dr. Neena Goveas

- Implemented a computational social psychology pipeline using fine-tuned Transformer-based emotion representations (RoBERTa) and directed network modeling to map peer support dynamics.
- Evaluated scale-free topology and tested emotion contagion across career-stress topics.
- Investigated empathy cascades in discussion threads to identify how context regulates the expression of peer support.
- Co-curricular engagement: Application of qualitative findings to design real time empathy interventions for the Gender equality cell on campus; coordinated and hosted outreach programs including interactive sessions — expression through fine arts, talks, films and music to create awareness regarding gender diversity among undergraduate and graduate students.

RESEARCH EXPERIENCE AND INTERNSHIPS

Stanford Medicine, Department of Pathology Jun 2025 – Aug 2025
Summer Research Intern (Khorana Scholar)
Supervisor: Dr. Le Cong

- Worked on CRISPR-based genome editing in mammalian systems for screening of cancer mutants.
- Data collection and VLM Performance scoring for LabOS:
  Cong, L., Smerkous, D., Wang, X., Yin, D., Zhang, Z., Jin, R., Wang, Y., Gerasimiuk, M., Dinesh, R. K., Smerkous, A., Shi, L., Zheng, J., Lam, I., Wu, X., Liu, S., Li, P., Zhu, Y., Zhao, N., Parakh, M., ... **Serrao, S.**, ... Wang, M. (2025). *LabOS: The AI-XR Co-Scientist That Sees and Works With Humans*. arXiv preprint arXiv:2510.14861. `https://arxiv.org/abs/2510.14861`

Unveiling Agency Dynamics of Older Adults in Care Homes in Goa, India, Managed by Churches and Government Bodies Feb 2024 - May 2024
Field Research Assistant, *Department of Humanities and Social Sciences - BITS Goa*

- Conducted bilingual (English and Konkani) ethnographic interviews with senior citizens in assisted living facilities to gather qualitative data. Coordinated fieldwork operations: the transcription of interview data and the identification of relevant government bodies and NGOs for institutional collaboration.

CSIR-Institute of Microbial Technology, Chandigarh May 2024 – Jul 2024
Summer Research Intern
Supervisor: Dr. Sanjeev Khosla

- Understanding the role of epigenetic modifications in host-pathogen interactions. Isolation of nuclei from Mammalian cells used to conduct MNase sequencing to understand chromatin dynamics and epigenetic regulation.

BITS Pilani, Dept. of Biological Sciences Aug 2024 – Jan 2025
Supervisor: Dr. Indrani Talukdar

- Analysis of Splice Variant Dataset of hnRNPD: Disease Mapping with Focus on Metabolic, Cardiovascular, and Muscular Disorders.

COURSEWORK AND SKILLS

Programming Languages: Python, Java, C, C++
Data Science/ML: PyTorch, HuggingFace Transformers, NetworkX, Pandas, NumPy, Scikit-learn
Spoken Languages: English, Hindi, Kannada, Konkani, Tulu
Selected Online Coursework (Coursera, 2024–2026): Social and Economic Networks: Models and Analysis (Stanford Online), Writing in the Sciences (Stanford Online), Introduction to Psychology, Moralities of Everyday Life, Narrative Economics (Yale), Soul Beliefs: Causes and Consequences (Rutgers), and Conspiracy Planet (UC Santa Cruz).
Humanities Electives: Introductory Psychology, Introduction to Philosophy, Comparative Indian Literature.

LEADERSHIP AND PUBLIC ENGAGEMENT

- President (2024–25) and Secretary (2023–24), Department of Biological Sciences, BITS Goa.
- Student President - Microbiologists Society of India, Goa (2026-2027).
- Member of the Commonwealth Student Community for AI Governance, India
- Member of the Equal Opportunities Committee (2024-2025); Anti-ragging Committee (2023-2024), BITS Pilani, KK Birla, Goa campus.
- Volunteer Teacher at Abhigyaan, BITS Pilani, KK Birla Goa Campus: Designed lesson plans and tutored underprivileged students from grades 1 through 12 as part of a student-run educational initiative.
- Core Member, Eco Club, BITS Goa: Coordinated student outreach for local sustainability programs, organizing beach clean-up drives, campus flora documentation (tree tagging), and nature walks.

INDEPENDENT PROJECTS

Project Resilience: Neuronal Transition-State Modeling, Independent Research Aug 2026 – Present

- An independent computational systems biology initiative to identify and model multi-omic molecular markers of transition states between neurological compensation and persistent dysfunction.
- Working with a team of medical and engineering students, analyzing complex interaction networks to identify dynamical network biomarkers to predict clinical deterioration prior to structural atrophy.

Project Kratos, Life Detection, Mars Rover Team, BITS Goa Sep 2023 – Sep 2025

- Employing chemical assays and environmental indicators to model the detection of extinct and extant life forms on the Martian surface; understanding the classification of rocks and terrains and their historical potential to support life during their formation process.
- Participated in the International Rover Challenge (IRC), 2024 held at PSG iTech, Coimbatore, India, as a member of Project Kratos, Life Detection Team.

MARS Learning Centre, Mangaluru Academy of Refined Studies Jun 2026 – Present

- Digital illustrator for Medical Education